

Compiler Construction (CS-4031)

Date: February 27th 2024

Course Instructor(s)

Dr. Faisal Aslam

Sessional-I Exam

Total Time: 1 Hours

Total Marks: 31

Total Questions: 08

Semester: SP-2024

Campus: Lahore

Dept: Computer Science

Student Name

Roll No

Section

Student Signature

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1. Create LL(1) parsing table for the following CFG. Must clearly show first and follow set of each non-terminal. [5 Marks]

$$\begin{aligned}
E &\rightarrow TE' \\
E' &\rightarrow +TE' \mid \varepsilon \\
T &\rightarrow FT' \\
T' &\rightarrow FT' \mid \varepsilon \\
F &\rightarrow id \mid (E)
\end{aligned}$$

$$\begin{aligned}
\text{First}(F) &= \{id, (\} \\
\text{First}(T') &= \{id, (\, \varepsilon\} \\
\text{First}(T) &= \{id, (\} \\
\text{First}(E') &= \{+, \varepsilon\} \\
\text{First}(E) &= \{id, (\}
\end{aligned}$$

$$\begin{aligned}
\text{Follow}(F) &= \text{First}(T') \cup \text{Follow}(T) \cup \text{Follow}(T') = \{id, (\, +, \$\} \\
\text{Follow}(T') &= \text{Follow}(T) = \{+, \$,)\} \\
\text{Follow}(T) &= \text{First}(E') \cup \text{Follow}(E) = \{+, \$,)\} \\
\text{Follow}(E') &= \text{Follow}(E) = \{ \$,)\} \\
\text{Follow}(E) &= \{ \$,)\}
\end{aligned}$$

	+	id	(	)	\$
E		TE'	TE'		
E'	$+TE'$			ε	ε
T		FT'	FT'		
T'	ε	FT'	FT'	ε	ε
F		id	(E)		

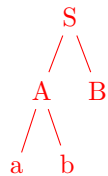
Table 1: Parsing Table

2. Given the following LL(1) parsing table, create parse tree for the input string **abde**. Must clearly show each stage of the stack, input consumed, and all the actions performed. [5 Marks]

	a	b	c	d	e	f	\$
S	$S \rightarrow AB$		$S \rightarrow AB$		$S \rightarrow eDa$		
A	$A \rightarrow ab$		$A \rightarrow c$				
B		$B \rightarrow dC$					
C					$C \rightarrow eC$		$C \rightarrow \epsilon$
D	$D \rightarrow \epsilon$					$D \rightarrow fD$	

Stack	inputs	Actions
S\$	abde\$	$S \rightarrow AB$
AB\$	abde\$	$A \rightarrow ab$
abB\$	abde\$	Match
bB\$	bde\$	Match
B\$	de\$	Reject

Partial Parse tree is as follows: (NOT COMPLETE)



3. Show that the following grammar is ambiguous for string $int--int-int$
[5 Marks]

$$\begin{aligned} S &\rightarrow E \\ E &\rightarrow E - E \\ &\quad | --E \\ &\quad | int \end{aligned}$$

I create two different parse trees for the string using the CFG as follows:

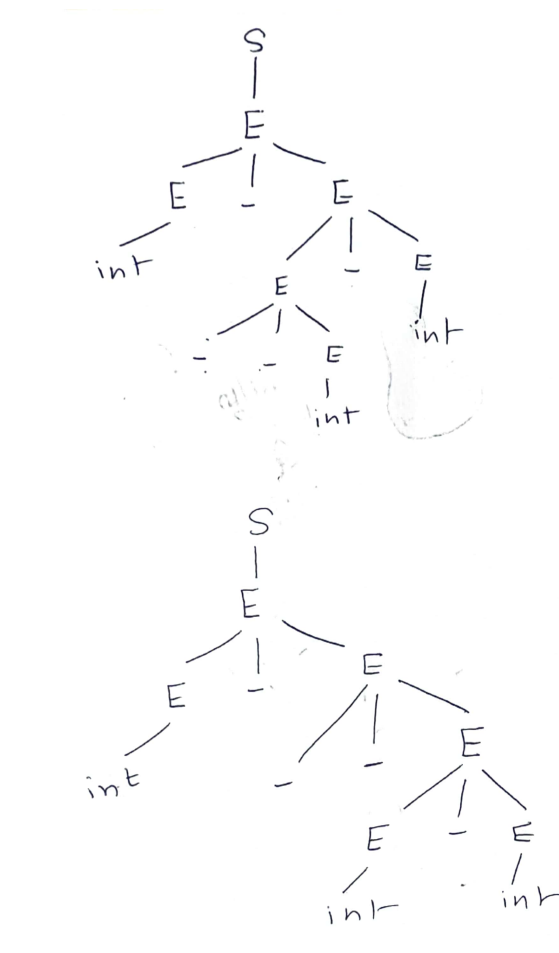


Figure 1: Two parse trees for the same input string

4. Perform left factoring on the following context-free grammar. [3 Marks]

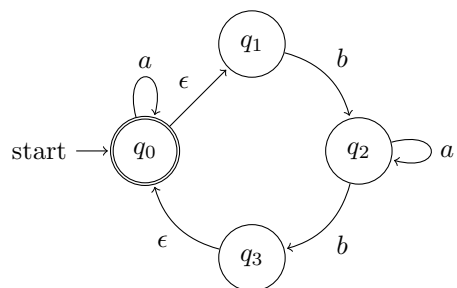
$$\begin{aligned} A &\rightarrow aBx \\ &\quad | aBy \\ B &\rightarrow cdg \\ &\quad | cd \end{aligned}$$

$$\begin{aligned} A &\rightarrow aBA' \\ A' &\rightarrow x \mid y \\ B &\rightarrow cdB' \\ B' &\rightarrow g \mid \varepsilon \end{aligned}$$

5. Change the following CFG to eliminate left recursion [3 Marks]

$$\begin{aligned} S &\rightarrow S \text{ and } S \\ &\quad | T \mid m \\ T &\rightarrow \text{true} \mid \text{false} \end{aligned}$$
$$\begin{aligned} S &\rightarrow TS' \mid mS' \\ S' &\rightarrow \text{and } S \ S' \mid \varepsilon \\ T &\rightarrow \text{true} \mid \text{false} \end{aligned}$$

6. Convert the following NFA to DFA. Please show all the steps. [5 Marks]



7. Write all the phases of compiler and input/output of each phase. [2 Marks]

Are not that obvious.

8. Write a CFG that starts with one or more y followed by twice as many x or z. Examples of string accepted by the grammar are: **yxx**, **yzz**, **yzx**, **yxx**, **yyxzzx**, **yyzxxz**, **yyyxxxxx**, ... [3 Marks]

$$\begin{aligned} S \rightarrow ySxx \mid ySzz \mid ySzx \mid ySxz \\ \mid yxx \mid yzz \mid yzx \mid yxz \end{aligned}$$



Course Name:	Compiler Construction	Course Code:	CS 4031
Program:	BS (CS)	Semester:	Spring 2023
Duration:	One hour	Total Marks:	30
Paper Date:	27-Feb-2023	Weight:	
Section:	6A and 8A	Page(s):	1
Exam Type:	Mid-1		

Student Name: _____ Roll No. _____

Solve question 1 on page 1, question 2 on page 2, and question 3 on page 3. Only the first three pages will be marked!

Question 1

Give token-lexeme pairs for the following source code:

```
int dif(int x, int y) {
    int d;
    if (x >= y)
        d = x - y;
    else
        d = y - x;
    return d;
}
```

$S \rightarrow S = T \mid T$
 $T \rightarrow S \mid F \mid S * F \mid F$
 $F \rightarrow id \mid num$

a =

Question 2

Give a DFA for the token *num*:

$num \rightarrow digit^+ F$

$F \rightarrow ^ \mid . digit^+ \mid n$

Halt the machine only after reading all the digits. You may use the keyword "other" and the symbol *

Question 3

Consider the following grammar

$S \rightarrow S \& S \mid S = S \mid S \# S \mid id \mid num$

Set precedence of operators & and # higher than that of = sign. Set associativity of all the operators from left to right (left associative).

$S \rightarrow S \& S \mid S = S \mid S \# S \mid id \mid num$
 $A \rightarrow A \& B \mid A = B \mid A \# B \mid id \mid num$
 $T \rightarrow id \mid num$



National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Compiler Construction	Course Code:	
	Degree Program:	BS (CS)	Semester:	Spring 2020
	Exam Duration:	60 Minutes	Total Marks:	30
	Paper Date:	Feb 2020	Weight	
	Section:	ALL	Page(s):	3
	Exam Type:	Mid-1		

Student : Name: _____ **Roll No.** _____

Section: _____

Instruction/ Attempt all questions. Programmable calculators are not allowed.

Notes:

Question 1 (5+5 marks)

a) Answer the following multiple choice questions:

1. A regular definition cannot contain a) Terminals b) Non-terminals c) Recursion d) Substitution rules (productions) e) Kleene star operator 2. Which of the following language processor provides code protection: a) Compiler b) Interpreter c) Hybrid system d) Both compiler and hybrid e) Both interpreter and hybrid 3. A symbol table contains information about a) Variables b) Functions c) Data types d) Identifiers e) Addresses	4. Which of the following language processor provides platform independence (portability): a) Compiler b) Interpreter c) Hybrid system d) Both compiler and hybrid e) Both interpreter and hybrid 5. Which of the following language processor uses both a compiler and an interpreter: f) Compiler g) Interpreter h) Hybrid system i) Both compiler and hybrid j) Both interpreter and hybrid
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b) Match the entries in the first column with the corresponding description in the second:

a) Optimization	i. Convert words into sentences
b) Parser	ii. Type checking
c) Semantic analyzer	iii. Partial evaluation of boolean expression
d) Short-circuit evaluation	iv. Execute program

e) Interpreter	v. Use less time and space
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Question 2 (5+5 marks)

a) Give a regular definition for a date. Following are few examples: 29-Feb-2020, 31-Mar-1982, 15-Dec-2010.

b) Give a DFA for odd-even language i.e. odd number of a's and even number of b's.

Question 3 (5+5 marks)

a) Give a CFG for the language: $\{a^n b^n \mid n \geq 0\}$.

b) Set precedence of the equality and inequality greater than the other two operators:

$E \rightarrow E < E \mid E > E \mid E == E \mid E != E \mid \text{num}$

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Compiler Construction	Course Code:	CS-402
	Program:	BS (CS)	Semester:	Spring 2018
	Duration:	60 Minutes	Total Marks:	35
	Paper Date:	28-Feb-2018	Weight	
	Section:	ALL	Page(s):	2
	Exam Type:	Midterm-I		

Student : Name: _____ **Roll No.** _____
Section: _____

Instruction/Notes: Solve question 1 on the question paper, while rest on the answer sheet.

Question 1 (5+5 marks)

a) Answer the following multiple choice questions:

1. Which of the following language processor provides fastest execution: a) Compiler b) Interpreter c) Hybrid system d) Both compiler and hybrid e) Both interpreter and hybrid 2. Which of the following language processor provides code protection: a) Compiler b) Interpreter c) Hybrid system d) Both compiler and hybrid e) Both interpreter and hybrid 3. Which of the following language processor is best for debugging: a) Compiler b) Interpreter c) Hybrid system d) Both compiler and hybrid e) Both interpreter and hybrid	4. Which of the following language processor provides platform independence (portability): a) Compiler b) Interpreter c) Hybrid system d) Both compiler and hybrid e) Both interpreter and hybrid 5. Which of the following language processor uses both a compiler and an interpreter: f) Compiler g) Interpreter h) Hybrid system i) Both compiler and hybrid j) Both interpreter and hybrid
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b) Match the entries in the first column with the corresponding description in the second:

a) Lexical analyzer	i. Convert words into sentences
b) Parser	ii. Type checking
c) Semantic analyzer	iii. Translator
d) Intermediate code generator	iv. Execute program
e) Interpreter	v. Identify words

Question 2 (5+5 marks)

a) Give output of a lexical analyzer for the following program:

```
int sum(int a[], const int N) {  
    int s = 0;  
    for (int i = 0; i < N; ++i)  
        s = s + a[i];  
    return s;  
}
```

b) Consider a programming language that allows programmer to write date-literal as follows:

Date birthday = 15-Apr-1990;

Here a new variable "birthday" of Date type is declared and initialized. The date literal has two integers, then three letters for the month, and finally four digits for year.

Give a regular definition for all possible date literals. You need not to perform any checks, such as the day cannot exceed 31, etc.

Question 3 (5+5+5 marks)

a) Consider the following grammar:

$E \rightarrow E + E \mid E - E \mid E * E \mid E / E \mid id \mid num \mid (E)$

Set precedence of addition and subtraction higher than that of multiplication and division. Also set associativity of all the operators from right to left.

b) Remove left recursion from the following grammar:

```
BE -> BE and BF | BE or BF | BF  
BF -> true | false | ( BE )
```

c) Left factor the following grammar:

```
S -> IF | ELSE  
IF -> if ( E ) S  
ELSE -> if ( E ) S else S
```

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Compiler	Course Code:	
	Degree Program:	BS (CS)	Semester:	Fall 2020
	Exam Duration:	90 min	Total Marks:	40
	Paper Date:	20-Oct-2020	Weight	
	Section:		Page(s):	4
	Exam Type:	Mid-1		

No separate answer sheets required. Rough sheet are allowed, however. Do not attach additional sheets with this question paper. All the questions carry equal marks

Question 1

α) Match the entries in the first column with the corresponding description in the second:

a) Lexical analyzer	i. Convert words into sentences
b) Parser	ii. Type checking
c) Semantic analyzer	iii. Translator
d) Intermediate code generator	iv. Execute program
e) Interpreter	v. Identify words

β) Answer the following MCQ's:

i. Compiler provides a) Fast execution b) Portability (platform independence) c) Better debugging d) Source code protection e) Both (a) and (d) ii. Interpreter provides a) Fast execution b) Portability c) Better debugging d) Source code protection e) Both (b) and (c) iii. Hybrid system provides a) Portability b) Better debugging c) Source code protection d) Better execution time e) All of above	iv. Following phases populate the symbol table a) Lexical analyzer b) Syntax analyzer c) Semantic Analyzer d) Both (a) and (b) e) Both (a) and (c) v. A regular expression cannot include a) Kleene star b) Operator or c) Concatenation d) Parenthesis e) Recursion
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Question 2

a) A cell phone user needs to dial a service code to access a particular service. Following are few example codes:

*234*6214532*500# *234*5274339*1000# *234*7318562*200#

Here, the code "234" is fixed. Afterwards there is a seven-digit phone number. Finally the user writes an amount between 100 and 1000. Only multiples of 100 are allowed: 572 is not allowed for example.

Give a regular definition for such service codes.

b) Give a DFA for the odd-odd language - odd number of a's and odd number of b's.

Question 3

a) Give a CFG for the C++ if-else statement. Note the else part is optional.

Hint: following is a CFG for the while loop:

$S \rightarrow \text{while} (E) S$

b) Set precedence and associativity in the following CFG:

$E \rightarrow E + E \mid E - E \mid E * E \mid E / E \mid \text{id} \mid \text{num}$

Assume the precedence of + is higher than all the other operators. Rest of the operators has the same precedence. Also assume the associativity is right to left.

Question 4

a) Remove left recursion from the following CFG:

$S \rightarrow S a \mid S b \mid x \mid y$

b) Remove left recursion from the following CFG:

$S \rightarrow E =: id$

$E \rightarrow num + E \mid num \mid S$

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Compiler Construction	Course Code:	
	Program:	BS (CS)	Semester:	Fall 2019
	Duration:	60 Minutes	Total Marks:	30
	Paper Date:	25-Sep-2019	Weight	
	Section:	ALL	Page(s):	3
	Exam Type:	Midterm-1		

Student : Name: _____ **Roll No.** _____
Section: _____

Question 1 (10 marks) Answer the following MCQ's:

i. Compiler provides a) Fast execution b) Portability (platform independence) c) Better debugging d) Source code protection e) Both (a) and (d)	vi. Lexical analyzer converts a) Characters into words b) Words into sentences c) Tokens into lexemes d) Lexemes into tokens e) Words into regular expressions
ii. Interpreter provides a) Fast execution b) Portability c) Better debugging d) Source code protection e) Both (b) and (c)	vii. Syntax analyzer converts a) Characters into words b) Words into sentences c) Parse trees into DFA's d) CFG into program e) PDA into CFG
iii. Hybrid system provides a) Portability b) Better debugging c) Source code protection d) Better execution time e) All of above	viii. Type checking is performed by a) Lexical analyzer b) Syntax analyzer c) Semantic analyzer d) Intermediate code generator e) Virtual machine
iv. Following phases populate the symbol table a) Lexical analyzer b) Syntax analyzer c) Semantic Analyzer d) Both (a) and (b) e) Both (a) and (c)	ix. Following phase captures syntax errors a) Lexical analyzer b) Syntax analyzer c) Semantic analyzer d) Intermediate code generator e) Virtual machine
v. Two different parse trees for a single input means a) Ambiguous grammar b) Ambiguous tree c) Efficiency d) Context free language e) Left recursion	x. A regular expression cannot include a) Kleene star b) Operator or c) Concatenation d) Parenthesis e) Recursion

Question 2 (5+5 marks)

a) Give regular definition for a date. A date shall consist of two digits, three-letters, and four digits, separated by dashes. For example 23-Sep-2019.

b) Give a DFA accepting the following language over $\{a, b\}$: The set of all strings whose fifth symbol from the right end is a "b".

Question 3 (5+5 marks)

- a) Rewrite the following grammar to set the precedence of "and" higher than that of "or". The precedence of parenthesis shall be highest.

$E \rightarrow E \text{ and } E \mid E \text{ or } E \mid (E) \mid \text{true} \mid \text{false}$

- b) Give an unambiguous CFG to generate all possible pairs of balanced parenthesis. Following are few examples:

$()$, $()()$, $()(())$, ...

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Compiler Construction	Course Code:	CS-402
	Program:	BS (CS)	Semester:	Fall 2018
	Duration:	60 Minutes	Total Marks:	30
	Paper Date:	2-Oct-2018	Weight	
	Section:	ALL	Page(s):	3
	Exam Type:	Midterm-I		

Student : Name: _____ **Roll No.** _____
Section: _____

Instruction/Notes: Solve all questions on this question paper. Do not attach any other answer sheet.

Question 1 (5+5 marks)

a) Fill in the blanks, using the following words: compiler, interpreter or hybrid system.

- _____ provides the fastest execution
- _____ does not hide source code
- _____ provides platform independence while protecting source code
- _____ is not good for debugging
- _____ uses both a compiler and an interpreter

b) Mark the following sentences as true or false:

- Lexical analyzer can determine the type of a variable
- Parser can check a function call for correct number of parameters
- Semantic analyzer performs type checking
- Intermediate code generator may generate three-address code
- Short-circuit evaluation is an error

Question 2 (5+5 marks)

a) Give output of a lexical analyzer for the following program:

```
int swap(int & x, int & y) {  
    int tmp = x;  
    x = y;  
    y = tmp;  
}
```

b) Consider a programming language that allows programmer to write time-literal as follows:

```
Time lunchTime = 01:30:00 pm;
```

Here a new variable "lunchTime" of Time type is declared and initialized. Your task is to give a regular definition for all valid time literals.

Question 3 (5+5 marks)

a) Consider the following grammar:

```
BE -> BE and BF | BE or BF | ( BE ) | BF
BF -> true | false
```

Rewrite the above CFG to set precedence of the logical operators. The precedence of "and" is higher than "or", while the precedence of parenthesis is highest.

b) Left factor the following grammar:

```
S -> id = E | single | compound
single -> while ( E ) S
compound -> while ( E ) { S }
```

Compiler Construction

First Sessional Exam, Fall 2016

Date: Sep 21, 2016

Marks: 40

Time: 60 min

Solve question 1 on page 1, question 2 on page 2, and so on.

Question 1 (5+5 marks)

- a) Give a regular definition for date constants, where each constant has a two-digit day, a three-lettered month, and a four-digit year. Following are few examples:

25 Feb 1875

03 Sep 1980

17 Jun 2016

You do not need to check for the days. For example, 35 Feb 1975 is incorrect, but you do not need to worry about such deeper issues.

- b) Give a transition diagram for time constants, where each constant has two-digit hours, minutes and seconds, separated by colons. Following are few examples:

09:10:20

15:30:40

22:50:55

Question 2 (5+5 marks)

- a) Set operator precedence in the following CFG as follows: "not" shall have highest, then comes "and", and the "or" shall have lowest precedence.

$BE \rightarrow BE \text{ and } BE \mid BE \text{ or } BE \mid \text{not } (BE) \mid \text{true} \mid \text{false}$

- b) Set the associativity of the "and" and "or" operators from right to left (right associative). Use the original grammar, given above.

Question 3 (5+5 marks)

- a) Remove any left recursion from the following grammar:

$L \rightarrow L , id \mid L : num \mid \# \mid *$

- b) Left factor the following grammar:

$D \rightarrow VD \mid FD$
 $VD \rightarrow T id ;$
 $FD \rightarrow T id (OPL) \{ L \}$
 $OPL \rightarrow PL \mid \wedge$
 $PL \rightarrow PL \mid T id \mid T id$

- Question 4 (10 marks)
Give any two important advantages of each of the following three language processors: Compilers, Interpreters, and Hybrid systems. Write only one or two lines for each advantage